● INCOMING CALL Mon Dec 7, 9:05 · Caller 4 · "The garbage can by the park on our corner is overflowing again." +0 codes picked while you read 00:00

SEM 620 · SOLUTION PROPOSAL · CLIENT: 311 AND CUSTOMER ACCESS

Same call. Same code. This time, the screen speaks up.

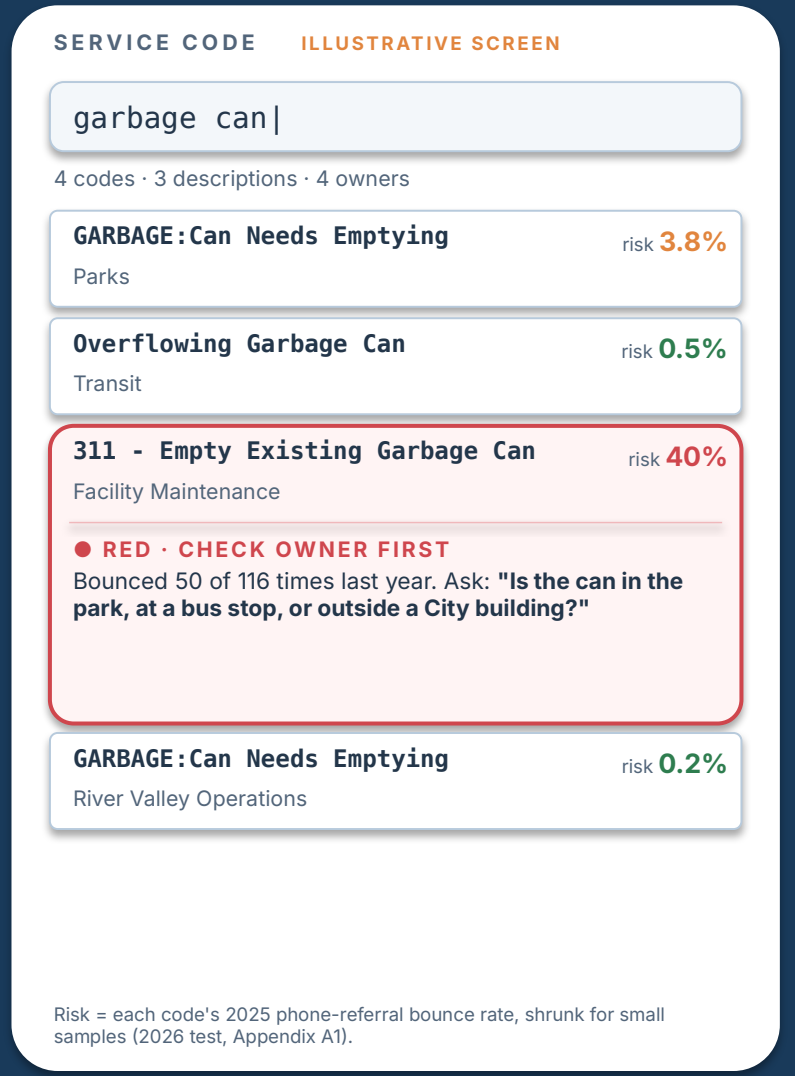
Route Check is a 20-second check on the **6% of phone referrals** most likely to land in the wrong area, and a loop that sends every miss back to someone who can fix it.

Tested on 2026 tickets it never saw, it flags 1 in 16 referrals and catches **41% of bounces**. The agent keeps the final pick.

Today: what it does, what data it needs, who has to change, and the first 90 days.

Rick Auger · Lauren Lonsberry · Nav Sekhon · Curtis Middleton

December 7, 2026



OUR RECOMMENDATION

Approve a **90-day start for Route Check** inside the current CRM, then decide on phase two by day 180 on evidence, not on belief

WHAT IT IS

A risk score on the code the agent picks, shown **only on risky codes**.

A daily bounce digest for team leads and a weekly code-fix queue with the owning areas.

It never routes a ticket by itself.

WHAT IT WILL DO

Catch about **4 in 10 bounces** before they happen, prompting on 1 in 16 referrals.[1]

Spare residents about **1,200 to 2,900 days** a year in the wrong queue.[3]

Build the routing record the 2030 CRM will need.[4]

WHAT IT WILL NOT DO

Save real money. City savings are about **$7K to $17K a year**, roughly the agent time it costs.[2]

Catch the other 6 in 10 bounces, or the misroutes areas fix quietly.

Replace agents, or judge them one by one.

WHAT WE ASK TODAY

Data Read access to the CRM's reassignment log and first-code field

People A 311 analyst, half-time, for 120 days; team leads for the digest

Partners CSU 52 and three area program coordinators in co-design

Decision An early look at day 120; go / no-go by day 180 against the gate on slide 13



LOGICAL LAUREN

The dollars are small, so we will not sell this as savings. We sell it as residents' days and a record the City does not have.



RULING RICK

Nothing here moves authority away from the agent or the areas. It gives the hand-off an owner.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026; Appendix A1-A2. 2 Using the October deck's decision tree ($1.02 per ask; asking halves wrong owners; $36.11 per bounce), annualised from 10.2 months of 2026. 3 Resident-days in the wrong queue avoided, range across tier settings. 4 CSD03148 (Sep 2, 2026): CRM end of life 2030.

THE CORE ANALYTIC DECISION · RESTATED AND REFINED

Our diagnosis moved: the pick is **usually right**. The gap is knowing **which picks are risky**, and hearing about the misses

OCTOBER

Which service code does a new 311 phone or web ticket get?

One agent, one pick, 1,917 codes, 665,225 times a year.

We showed that a wrong code costs residents days, and that nothing tells the agent.

DECEMBER · TWO LINKED DECISIONS

1 PER TICKET · LIVE · THE AGENT

Route on the code I picked, or check who owns it first?

Asking pays only when a code's bounce chance is above 5.6% (the October decision tree). Today nothing tells the agent that chance.

2 PER CODE · WEEKLY · 311 WITH THE OWNING AREA

Keep, reword, merge or retire this code?

A code that keeps bouncing is a code-set problem, not an agent problem. Today no one owns that fix.

98.9%
of 2026 phone referrals did not bounce

37%
of 2025 phone-referral bounces sit on just 10 codes

233 of 1,116
referral codes bounced at all in 2025

50 of 116
bounced on the picked can code, 2025



LOGICAL LAUREN · RATIONAL

Concentrated misses mean a targeted check beats a check on every call.



NEURO NAV · PSYCHOLOGICAL

Agents can't learn from a miss they never hear about.



RULING RICK · POLITICAL

311 picks the code; areas own the result. Decision 2 gives the hand-off an owner.

1 October deck, slide 4. 2 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: 2026 phone referrals Jan 1-Nov 6 (179,139; 1,930 bounces); 2025 phone-referral bounces by code. 3 Glaser, Analytic Trees p. 20 (threshold); October deck slide 11. 4 Glaser, The Psychological Perspective 3.3; The Reassembly of Agency 1.1.

THE PROPOSED ALGORITHMIC ROUTINE · ROUTE CHECK

Route Check **takes in the agent's pick**, puts out a prompt only when the pick is risky, and **sends every miss back** to someone who can fix it

TAKES IN

The code the agent picked
on every phone referral

Code risk table
12-month bounce rate per code, weekly[1]

Sibling map
look-alike codes owned elsewhere[2]

Phase two: address and notes
asset layers; agent's typed words

SCORES

A lookup, not a black box

```
risk = (bounces + prior) ÷ (referrals + prior)
```

Explainable in one sentence: **"This code bounced 50 of 116 times last year."**

New codes start amber for 90 days.

PUTS OUT

GREEN · under 2% ~94% of referrals
Nothing shown. Route as today.

AMBER · 2% to 5.6% ~5%
Show the look-alike codes and who owns each. One click to confirm or switch.

RED · above 5.6% ~1%
Ask one ownership question, then pick.

WHO ACTS

The agent

hears the caller, answers the prompt, keeps the final pick. Overrides take one click and are logged.

No auto-routing

Only the agent hears the caller. A wrong automatic route would have no one to catch it.

THE LOOP · WHAT HAPPENS AFTER A MISS

Fills the two gaps from October: AUTHORIZE and LEARN[3]

DAY 0-3
A bounce
The receiving area closes it "311 Redirecting"

NEXT DAY
Daily digest
Team lead sees their team's bounces; coaching only

WEEKLY
Weekly fix queue
311 and the owning area keep, reword, merge or retire codes above 5.6%

WEEKLY
Weekly refresh
311 analytics recomputes the risk table; watches drift

1 Risk table: each code's bounce rate over the last 12 months, shrunk toward the average for small samples (empirical Bayes); refreshed weekly. 2 Sibling map: codes with overlapping wording owned by another area; draft of 301 codes from 2025 (sibling_codes_2025.csv). 3 Six activities: Glaser, The Reassembly of Agency s. 1.3.

WHAT THE AGENT SEES · THREE TIERS

The prompt **interrupts the fast pick only where it is likely wrong**: nothing on 94% of referrals, one click or one question on the rest

GREEN Risk under 2%

93.6%

of 2026 phone referrals

Nothing appears. The agent routes as today, at today's speed.

EXAMPLE Prune City Tree · Forestry · 1.7%

AMBER Risk 2% to 5.6%

5.2%

of referrals · 24% of bounces

The look-alike codes appear with their owners. One click confirms or switches.

EXAMPLE LITTER:Road · Roadway · 5.0% → siblings shown

RED Risk above 5.6%

1.2%

of referrals · 18% of bounces

One ownership question appears. The agent asks, then picks.

EXAMPLE "Is it in the park, at a bus stop, or by a City building?"

RULES THAT KEEP IT HONEST

Cut-offs come from the analysis, not taste: red starts at 5.6%, the October decision tree's break-even.[2]

New codes start amber for 90 days: 8% of 2026 bounces came from codes unseen in 2025.

Small samples cap at amber: one bounce on five tickets should not trigger a question.[4]



NEURO NAV · PSYCHOLOGICAL

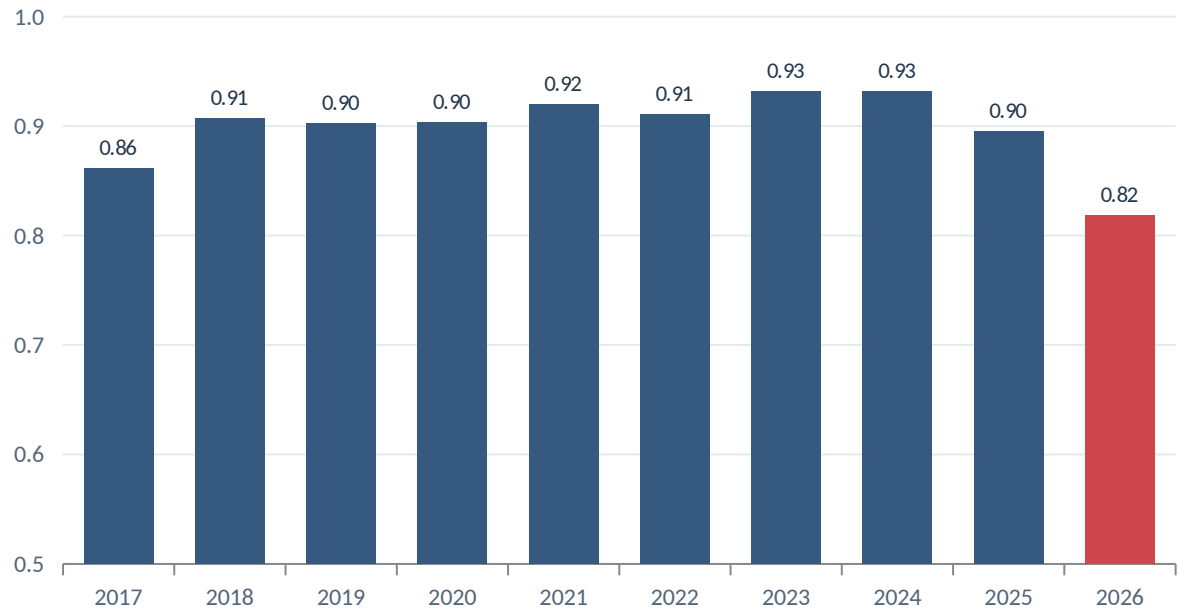
A prompt on every call becomes wallpaper and I stop reading it. A prompt on one call in sixteen tells me something is different about this one. That's when my slow thinking should switch on.[3]

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: tier shares of 2026 phone referrals using 2025 risk scores. 2 October deck slide 11: asking pays when p > 2 x $1.02 / $36.11 = 5.6%. 3 Glaser, The Psychological Perspective 3.2. 4 Our proposed refinement, to test in shadow mode: 61 of 91 red codes had fewer than 20 referrals in 2025.

DOES IT WORK? · BACKTEST ON TICKETS THE SCORE NEVER SAW

Last year's code bounce rates rank next year's tickets well every year since 2017; a fancier model did worse

HOW WELL THE SCORE RANKS NEXT YEAR'S TICKETS (AUC)[1]



2026 is the weakest year because eight areas began recording bounces; 0.82 is still strong. Three years of history did no better than one.

WHERE TO SET THE FLAG · 2026 TEST

Flag codes with risk	Referrals flagged	Bounces caught	Bounce rate if flagged
above 5.6% (red only)	1.2%	18%	16.2%
above 2% (red + amber)	**6.4%**	**41%**	**7.0%**
above 1%	12.0%	50%	4.4%
no flag (today)	0%	0%	1.08% overall

WHAT WE TRIED THAT DID NOT HELP

More features: a regression adding app bounce rate, area and sibling flags scored AUC 0.69 out of time, worse than the lookup's 0.82.

Neighbourhood: predicting the can owner from neighbourhood was right 62.8% of the time vs 63.9% for always guessing the biggest area.

Longer history: three years of data matched one year within 0.01 AUC.



LOGICAL LAUREN · RATIONAL

41% caught means 59% missed. Route Check is triage, not a cure. That is why the loop on slide 4 matters as much as the prompt.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026; 2016-2022 from the City's yearly 311 datasets (deduplicated by Row ID). Each year scored only with the year before (out-of-time test). AUC: chance a random bounced ticket scores above a random non-bounced one (0.5 = coin flip). 2 Appendix A2.

WHAT WE AUTOMATE · WHAT WE LEAVE TO PEOPLE · AND WHY

Automate **the counting and the reminding**; leave **the pick, the code fixes and the coaching** to people who hear the caller and own the outcome

Automate	Leave to a person	Why
Score every pick; choose the tier; show look-alike codes	The final code (agent)	Only the agent hears the caller. An automatic wrong route has no one to catch it, and Hashemi (2022) shows text models still miss about 1 in 6.
Build the daily digest and the weekly fix queue	Retiring, merging or rewording codes (311 + area)	Areas already sign off scripts. Changing the code set moves work between budgets: a political choice, not a statistical one.
Weekly refresh; drift and recording alerts	Coaching (team lead)	Feedback changes behaviour when it comes from a person the agent trusts, not from a dashboard.
Log every override with the outcome	Whether the pilot continues (311 director)	Someone accountable must be able to stop it.

THE FOUR QUESTIONS FOR ANY AGENT[1]

ACCESS — Agents see why a code is flagged; team leads see only their own team's bounces.

CHANGE — Cut-offs and the sibling map change only through the weekly fix queue, with area sign-off.

STOP — The 311 director can switch it off; it switches off by itself if handle time rises more than 10 s.

CORRECT — One-click override, logged; overrides that turn out right lower the code's score.



RULING RICK · POLITICAL

Northfield's override was allowed on paper, but a throughput metric made it costly. We judge handle time by team, never by agent, and no one is marked down for an override.[2]



NEURO NAV · PSYCHOLOGICAL

When Ironstone replaced Wes Harlan, his tacit calls went with him. Agents know that cans at transit centres are ETS. The override log and fix queue capture that.[3]



LOGICAL LAUREN · RATIONAL

Route Check is an analyst, not a decider: it adds evidence to the agent's pick. Phase two adds an assistant that suggests three codes.[4]

1 Glaser, The Reassembly of Agency (SEM 620, 2026), s. 1.1 and Session 1: who can access, change, stop and correct an agent. 2 Northfield Credit Union case (Session 1): a score with a 1-in-20 override, judged by a throughput metric. 3 Ironstone case (Session 7). 4 Glaser, Sloan and Knight (2025): AI as assistant, analyst, coach.

WHEN IT IS WRONG · A PREMORTEM[2]

It is December 2027 and Route Check has failed. Here are the six likeliest reasons, and what we built in for each

FALSE ALARM

A green-tier pick gets flagged

Costs about 20 seconds. One-click override; the override and its outcome feed the next refresh.

WATCH Override rate by code

MISSED BOUNCE

6 in 10 bounces are on unflagged codes

Same as today, except the bounce now reaches the team lead next day and the code enters the fix queue.

WATCH Bounces returned within 24 h

DRIFT

An area starts or stops recording bounces[1]

Track the recording rate per area; judge the pilot only on areas that always recorded.

WATCH Recording rate by area

RUBBER-STAMPING

Agents click through every prompt[4]

If agents accept more than 95%, audit a sample. If they override more than 80%, the tier is noise: retune it.

WATCH Override rate 20% to 80%

SURVEILLANCE

The digest becomes a ranking of agents

Written agreement with CSU 52: team-level data, coaching only, no individual targets.

WATCH Grievances; agent survey

PROXY DRIFT

Areas stop recording bounces to look good[3]

Never make bounces an area target. Pair them with a QA sample of 200 resolved calls a month and a re-route log.

WATCH QA sample vs bounce trend



RULING RICK · POLITICAL

The most dangerous failure is the quiet one: once a bounce count is on a scorecard, someone has a reason to stop recording bounces. Robodebt is what a trusted proxy looks like when no one checks it.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: 39% of the 2026 rise in phone bounces came from eight areas that began recording (October deck slide 7). 2 Klein (2007), Performing a project premortem, HBR. 3 Lindebaum et al. (2022), as assigned in SEM 620 Session 9; Robodebt simulation. 4 Parasuraman and Manzey (2010), Human Factors 52(3).

HOW DATA WOULD BE CONSTRUCTED AND LEVERAGED

Public data is enough to build version one. The learning loop needs **five records nobody keeps yet**, and small new habits from agents, area staff and 311

Data	Status today	Used for	Who has to change
Code, area, channel, closure label, created and closed times	EXISTS · public[1]	Risk table, backtest, tiers	No one
Ticket address, agent ID, handle time, typed notes	EXISTS · inside CRM[2]	Digest by team; phase-two owner hints and text model	311: a data agreement
Parks and transit stop locations	EXISTS · public[4]	Red-tier question: whose asset is at this address?	No one
Link from a bounced ticket to its re-route (first code, final code)	MISSING · cannot rebuild[3]	The label every later model learns from	Area staff: log the code it should have been
A closure reason recorded the same way in every area	PARTIAL · 8 areas began 2026	A fair comparison across areas	Area staff: one required field
Sibling map of look-alike codes	DRAFT · 301 codes	Amber prompt	311 knowledge staff and areas confirm it
Code accuracy on calls resolved on the phone (473,564 in 2025)	MISSING	The stake's biggest assumption	311 QA: code a sample of 200 a month
Override log: prompt shown, agent's choice, outcome	MISSING · comes with Route Check	Trust, drift, retuning	Agents: one click

DATA IS BUILT, NOT FOUND

Today the only routing signal is a closure label set by the receiving area, and only when it chooses to use it.

The City's headline, 98.7% resolved on initial contact, measures the call, not the route: the same trap as the 10-1-24 reply target in the SBS case.

LEVERAGE

Every row in red is also a requirement for the 2030 CRM. Building them now gives the new system three years of clean routing labels on day one.



NEURO NAV · PSYCHOLOGICAL

The re-route link is the one that matters most: it turns a bounce from "not mine" into "it should have been this code", which is something an agent, and a model, can learn from.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: 26 columns; no address, no agent, no free text; neighbourhood centroid only, missing on 38% of 2026 phone referrals. 2 Inferred from OCA (2023) and CSD03148; to confirm with 311. 3 Matching bounces to a later ticket nearby found 19.3% vs 16.8% by chance. 4 Parks (gdd9-eqv9), ETS GTFS stops (4vt2-8zrq).

THE BUSINESS CASE · HONEST ABOUT THE DOLLARS

In City dollars Route Check is **roughly break-even**. It is worth doing for residents' time, hazard tickets, and the 2030 CRM

CITY DOLLARS · A YEAR[1]

$7K to $17K
gross savings from fewer bounces

$3K to $14K
agent time spent on prompts

RESIDENTS · A YEAR

1,200 to 2,900
days not spent in the wrong queue

2,915 days
hazard tickets sat misrouted in 2025: roads, sidewalks, signals[2]

THE 2030 CRM · OPTION VALUE

3 years
of routing labels ready when the CRM is replaced[3]

Waiting is not free: every year without a re-route link is a year of labels the new system can never train on.

WORKING BACK FROM 2030 · FUTURE → NOW

2026-27
Route Check pilot; re-route log and closure reason in the busiest areas

2027-28
All teams; text suggestions on agent notes; routing measure reported to Council

2029
CRM replacement specification written from three years of logged routing data

2030
New CRM routes with tested, City-owned data and an agent override already trusted



LOGICAL LAUREN · RATIONAL

Put the potential loss beside the average: a 0.5% bounce rate hides a 40% code, and 9% of bounces waited over a month.



RULING RICK · POLITICAL

Council asked about closures and duplicates. A routing measure next to 98.7% answers a question they will ask next.

1 Using the October decision tree: $1.02 per 20-second ask; asking halves wrong owners; $36.11 per bounce incl. expected field visits; annualised from 10.2 months of 2026. 2 October deck slide 14: 231 hazard-type phone bounces, 2,915 days in the wrong queue, 2025. Alberta MGA s. 532. 3 CSD03148 (Sep 2, 2026).

INTEGRATION · THE POLITICAL GAME[1]

Every player gets something they already care about and **no one loses authority**; the union and the areas join on day one

Player	Measured on today	What they fear	What Route Check gives them	Our move
311 agents (CSU 52)	Answer speed; resolved on first contact	Being replaced or ranked	Fewer repeat calls; a reason behind each prompt; final say	Co-design prompts; written coaching-only agreement
311 team leads	Team service levels	More reporting work	A daily digest that does the counting	Digest fits in a 10-minute huddle
311 director	98.7% resolved on initial contact; 120 s answer	Longer calls; a new bad-news number	A routing measure they control before Council asks	Guardrail: +10 s handle time stops the pilot
Area program coordinators[2]	Their own timeliness; script sign-off	Blame for bounces; more tickets	Fewer wrong tickets in their queue; a say in the code set	Co-own the fix queue; no area bounce target
Council / City Manager	Complaints; duplicate tickets; audits	Another IT project	Evidence before money; a CRM requirement list	Report the measure next to 98.7% from phase two



RULING RICK · POLITICAL

Authority over the code stays with 311 and accountability for the work stays with the areas.[3] What changes is that a miss now has an owner: the weekly fix queue, run by both.



NEURO NAV · PSYCHOLOGICAL

If agents hear "AI" and "accuracy" first, they'll hear "replacement". Start with the can code they already know bounces, and with their own override stories.

1 Allison and Zelikow (1999), as used in the Session 4 debrief: where you stand depends on where you sit. 2 OCA (2023) p. 20: program coordinators sign off script content. 3 CSD03148 p. 2: business areas assume full ownership after routing. Stakes are our hypotheses, to test in co-design (not interviews held).

PHASED IMPLEMENTATION · THE FIRST 90 DAYS

Fix the worst codes first, run in **shadow mode** before anyone sees a prompt, then pilot on half the floor **against a control**

DAYS 1-30
Fix and build
No model on screen

- Data agreement; pull the CRM reassignment log and agent IDs
- Re-run the backtest on internal data
- Quick win: review the two Facility Maintenance can codes with the areas: 8% of 2025 phone-referral bounces[1]
- Co-design prompts with agents and CSU 52; sign the coaching-only agreement

DAYS 31-60
Shadow mode
Score all, show none

- Check that the flagged share matches the backtest (about 6%)
- Start the daily digest to team leads
- Start the weekly fix queue with three areas: Roadway, Parks, Facility Maintenance
- Areas add a required closure reason and "should have been" code

DAYS 61-90
Pilot
Prompts on, half the teams

- Amber and red prompts on for half the teams; the rest are the control[2]
- Monthly QA sample of 200 resolved calls begins
- Weekly guardrail check: handle time, override rate
- Pilot runs to day 180 unless the day-120 look settles it[3]

DAY 120 EARLY LOOK · DAY 180 GO / NO-GO Two months of pilot can show a 40% drop; a 30% drop needs about 3.6 months. At day 120 we stop for harm or go early on a big effect. Otherwise the gate meets at day 180, and a no-go is a real option.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: "311 - Empty Existing Garbage Can" 50 + "311 - Request New Garbage Can" 35 = 85 of 1,062 phone-referral bounces in 2025. 2 Randomised by team-week so the comparison is fair. 3 Appendix A4: about 1,100 flagged tickets a month.

WHAT HAS TO BE TRUE BEFORE PHASE TWO · HOW THE CITY KNOWS IT WORKS

Phase two starts only if bounces fall on flagged codes **without slowing calls**, agents trust the prompt, and the areas are logging re-routes

THE GATE · ALL SIX BY DAY 180

1. Bounce rate on flagged codes down 30% or more vs control
2. Average handle time up 10 seconds or less; 120 s answer target still met
3. Override rate between 20% and 80%
4. Re-route log live in areas receiving 80% of referrals
5. Text model reaches 85% top-three accuracy on agent notes, offline[3]
6. CSU 52 and area sign-off to expand

THE SCORECARD[1]

Measure	Baseline	Target	Type
Bounce rate on flagged codes, pilot vs control	7.0%	down 30%+	Primary
Median days in the wrong queue per bounce	2.0 days	down	Primary · residents
Bounces returned to the team within 24 hours	none today	90%	Learning loop
Codes cleared from the fix queue per quarter	none today	rising	Learning loop
Average handle time	today	+10 s or less	Guardrail
Override rate	none today	20% to 80%	Guardrail
Bounce recording rate by area	varies	stable	Data health

Do not judge it on the City-wide bounce rate. Weekly rates swing from 0.79% to 2.05% with season and recording; that noise would swamp the effect.[2]

AFTER THE GATE Phase two (months 7-12): all teams; address-to-asset owner hints on red; text suggestions of the top three codes from the agent's notes, reusing the app's July 2026 AI, still a suggestion. **Phase three (2028-30):** routing measure reported to Council; data requirements written into the CRM replacement.

1 311 Requests open data (q7ua-agfg), pulled Nov 7, 2026: flagged-code bounce rate 7.0% in 2026; median 2.0 days per phone bounce, 2026. 2 Weekly phone-referral bounce rate ran 0.79% to 2.05% between April and October 2026. 3 Hashemi (2022) reports over 83% top-one accuracy on 311 descriptions; our 85% top-three gate is our assumption.

THE SHIFT DEBRIEF · WHAT WE PROPOSE

One check on the risky 6%, one loop for every miss, and the agent still makes the call

MON 9:06
The can code turns red. The agent asks: park, bus stop or City building?

MON 9:07
"The bus shelter." The agent switches to the Transit code.

TUE
Transit empties the can. Ticket closed, resolved.

THU
No second call. The can code is on the fix queue.



LOGICAL LAUREN

It costs about what it saves in City dollars. It is worth it for residents' days and for the data the 2030 CRM needs.



NEURO NAV

Agents finally hear about their misses, the next day, from a person they trust.



RULING RICK

Nobody loses authority. A miss now has an owner, and the union is in the room.

DECISION REQUESTED TODAY

Approve days 1-30: a data agreement, a half-time analyst, and co-design with CSU 52 and three areas. **Nothing reaches an agent's screen until shadow mode confirms the backtest.**

Illustrative story; codes and rates real (311 Requests open data, 2025). Sources: Appendix A6.

APPENDIX A1 · METHOD

The score is one ratio per code, shrunk so rare codes are not judged on one or two tickets

DEFINITIONS

Bounce: a ticket closed "311 Redirecting to Other Business Area".

Code: service area + category + description. We count 1,930 phone codes in 2025; the October deck counted 1,917 (definition difference, e.g. blank descriptions).

Population: phone tickets of type Referral, where 94% of phone bounces sit.

Score: (bounces + a) ÷ (referrals + a + b); prior mean 0.51%, weight about 10 tickets.

Test: fit on 2025, test on Jan 1-Nov 6, 2026 only; repeated for every year 2017-2026 using the prior year.

Data: 3,531,392 tickets (2023 to Nov 6, 2026) from the live dataset; 2016-2022 from the City's yearly 311 files, deduplicated by Row ID (the 2017 file lists most rows twice).

KNOWN WEAKNESS WE FOUND IN REVIEW

A prior weight of about 10 tickets is light: a code with 1 bounce on 5 tickets scores 7.2%, which is red. 61 of the 91 red codes had fewer than 20 referrals in 2025.

Fix to test in shadow mode: cap codes under 20 referrals at amber, or raise the prior weight. The 91 red codes together carry 2,579 referrals and 395 bounces (15%).

WHAT THE LABEL CANNOT SEE

Quiet misroutes fixed inside an area, and wrong answers on 473,564 phone-resolved calls (2025), leave no record. The score learns only from recorded bounces.

Recording changes move the label: all-channel bounces jumped from 2,162 (2021) to 4,515 (2022) and again in 2026 as more areas recorded (17 areas in 2021, 31 in 2026).

311 Requests open data (q7ua-agfg), pulled Nov 7, 2026; scripts in edmonton-311-analysis/scripts (load.py, a1-a4.py, h1.py). Shrinkage: empirical Bayes with a beta prior fitted to 2025 phone referrals.

APPENDIX A2 · STABILITY, 2016-2026

The score has held for ten years; the bounce rate stepped up twice, both times when more areas began recording

Year	Phone tickets	Phone referrals	Phone-referral bounces	Bounce rate	Areas recording	Phone codes	AUC (prior year)	Caught at 2% flag
2016	1,831,720	207,617	580	0.28%	17	1,701	--	--
2017	1,832,041	259,028	585	0.23%	20	1,738	0.86	28%
2018	1,691,830	261,243	649	0.25%	14	1,766	0.91	35%
2019	1,470,990	254,693	566	0.22%	19	1,828	0.90	44%
2020	935,051	200,313	562	0.28%	17	1,881	0.90	36%
2021	938,844	225,435	584	0.26%	17	1,990	0.92	33%
2022	816,788	223,944	1,205	0.54%	20	1,967	0.91	43%
2023	734,419	193,077	1,062	0.55%	23	2,020	0.93	58%
2024	704,535	188,003	874	0.46%	20	2,025	0.93	64%
2025	665,227	191,521	1,062	0.55%	24	1,930	0.90	56%
2026*	557,353	179,139	1,930	1.08%	31	1,954	0.82	41%

* 2026 to Nov 6. Phone volume fell by about two thirds from 2016 to 2025 while phone codes grew from 1,701 to 1,930: fewer calls spread over more codes means less practice per code.

311 Requests open data (q7ua-agfg), pulled Nov 7, 2026; yearly_2016_2026.csv and stability_backtest_2017_2026.csv in edmonton-311-analysis/.

APPENDIX A3 · TOP CODES BY 2025 PHONE-REFERRAL BOUNCES

Ten codes carry 37% of bounces; three of the top five belong to Facility Maintenance

Area	Description	Referrals	Bounces	Risk score	Tier
Roadway Maintenance	LITTER:Road	2,089	104	5.0%	AMBER
Roadway Maintenance	Callback Request	1,655	57	3.4%	AMBER
Facility Maintenance	311 - Empty Existing Garbage Can	116	50	39.8%	RED
Facility Maintenance	311 - Request New Garbage Can	59	35	51.1%	RED
Facility Maintenance	311 Parking Lot	135	30	20.8%	RED
Parks	GENERAL COMMENT: COE Parks Services	392	27	6.7%	RED
Traffic	General Comments Traffic Services	1,802	27	1.5%	GREEN
Roadway Maintenance	General Comments Roadway Maintenance	679	26	3.8%	AMBER
Forestry	Prune City Tree	1,069	18	1.7%	GREEN
Roadway Maintenance	LITTER:Boulevard Litter	408	17	4.1%	AMBER
Parks	GARBAGE:Other	267	16	5.8%	RED
Parks	SIGNS:Sign Required	34	14	32.2%	RED

READING THE TABLE

Two kinds of problem show up.

Look-alike codes (the can codes, parking lot): the wording fits several owners. Fix in the code set and with a red question.

Catch-all codes (General Comments, Callback Request): high volume, low rate, used when nothing else fits. Fix by splitting or scripting, through the fix queue.

Tiers: 956 green, 68 amber, 91 red codes in 2025.

311 Requests open data (q7ua-agfg), pulled Nov 7, 2026; code_risk_table_2025.csv. Tier from the shrunk 2025 rate.

APPENDIX A4 · PILOT POWER AND MEASUREMENT

A 30% drop needs about 3.6 months of pilot to show, so the gate sits near day 180; the City-wide rate is too noisy to judge on

Effect to detect	Months of pilot needed	Implication
50% drop (7.0% → 3.5%)	about 1.2	Visible by day 120
40% drop (7.0% → 4.2%)	about 1.9	Day-120 early look
30% drop (7.0% → 4.9%)	about 3.6	Full gate near day 180
Under 20%	much longer	Treat as no effect

WHY RANDOMISE BY TEAM-WEEK

Seasons and recording changes move bounce rates week to week. Splitting teams at random, and switching weekly, keeps both arms exposed to the same weather, areas and codes.

WHY NOT THE CITY-WIDE BOUNCE RATE

Weekly phone-referral bounce rates ran from 0.79% to 2.05% between April and October 2026 (mean 1.33%). Route Check works on 6% of referrals; even a large effect on them moves the City-wide rate by less than the week-to-week noise. **Judge it on the flagged codes, against the control.**

Two-proportion test, 80% power, 5% two-sided significance, baseline 7.0% bounce rate in the flagged set, about 1,100 flagged tickets a month split 50/50; prompts live from day 61. Team-week randomisation adds a design effect: treat as minimums.

APPENDIX A5 · COURSE CONCEPTS

Every design choice traces to a concept we used in October, so the solution follows from the diagnosis

Design choice	Concept	Session / source	Slide
Red tier at 5.6%	Decision tree threshold, solved right to left	S2 · Analytic Trees p. 20	5
Shrunk per-code rates; flagged-code measure	Look inside the average; mix effect	S2 · Analytic Trees p. 13	6, 13
Prompt on only 6% of picks	System 1 / System 2; automation bias	S3 · Psych. Perspective 3.2; Parasuraman & Manzey	5
Daily digest, next day	Intuition needs timely feedback	S3 · 3.3; Kahneman & Klein 2009	4
Agent keeps the final pick; four questions	Reassembly of agency; access, change, stop, correct	S1 · Reassembly 1.1, 1.7	7
Northfield guardrail; Ironstone override log	Score + override; tacit knowledge lost to systems	S1, S7 cases	7
Route Check as artifact; fix queue as routine	Ostensive, performative, artifact	S7 · Feldman & Pentland 2003	4
Fix queue co-owned with areas	Political game; where you stand depends on where you sit	S4 · Allison	11
Five missing records; re-route link	Data is constructed; metric built slightly wrong	S6 · SBS Ivey case	9
No area bounce target; QA sample	Proxy drift; values withering under algorithms	S9 · Robodebt; Lindebaum et al.	8
Premortem of six failures	Premortem	S2 · Klein 2007	8
Working back from the 2030 CRM	Strategic foresight (Future → Now); option value	S4, S2	10

SEM 620 readings and cases (Glaser 2026); Feldman and Pentland (2003) ASQ 48(1); Kahneman and Klein (2009); Klein (2007); Parasuraman and Manzey (2010); Lindebaum et al. (2022); Glaser, Sloan and Gehman (2024); Glaser, Sloan and Knight (2025).

APPENDIX A6 · AI PROCESS (SUMMARY) AND SOURCES

AI pulled and tested 3.5 million tickets in hours; people chose the decision, caught its errors, and own every claim

AI PROCESS · SUMMARY

Tools: Claude (Anthropic) in Claude Code projects: separate threads for data analysis, course-concept reference, and deck drafting.

What AI did well: pulled the full open dataset, ran out-of-time backtests, and told us the dollar case was weak.

What review caught: a second AI pass found light shrinkage making one-bounce codes red, and a code-count mismatch (1,930 vs 1,917). Team to verify.

What AI could not do: talk to 311 agents or area staff. Every stake on slide 11 is a hypothesis.

Rules: public data only; no City or personal data entered; the team rewrites all final text before submission.

SOURCES

City of Edmonton. 311 Requests open data (q7ua-agfg). data.edmonton.ca/d/q7ua-agfg, accessed Nov 7, 2026; yearly 311 datasets 2016-2022.

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Brynjolfsson, E., Li, D., & Raymond, L. (2023). Generative AI at Work. NBER w31161. nber.org/papers/w31161

Glaser, V. L. (2026). The Reassembly of Agency; The Psychological Perspective; Analytic Trees; SEM 620 session slides and cases.

Alberta, Municipal Government Act, RSA 2000, c M-26, s. 532. October 26, 2026 team deck, Appendix H, for all other sources.

Full AI process reflection submitted separately.